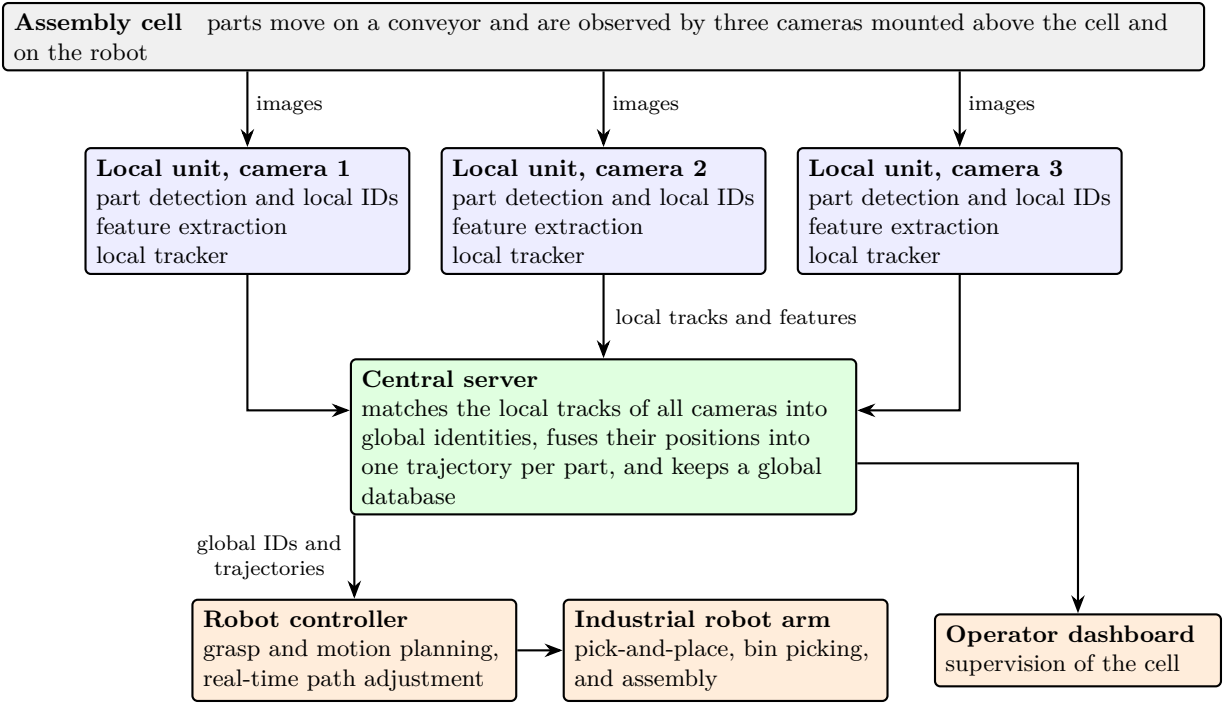

Assembly cell parts move on a conveyor and are observed by three cameras mounted above the cell and on the robot
images
images
images
Local unit, camera 1
part detection and local IDs
feature extraction
local tracker
Local unit, camera 2
part detection and local IDs
feature extraction
local tracker
Local unit, camera 3
part detection and local IDs
feature extraction
local tracker
local tracks and features
Central server
matches the local tracks of all cameras into global identities, fuses their positions into one trajectory per part, and keeps a global database
global IDs and trajectories
Robot controller
grasp and motion planning, real-time path adjustment
Industrial robot arm
pick-and-place, bin picking, and assembly
Operator dashboard
supervision of the cell